PROJECT REPORT

Mid-Mesh Generation of a Plastic Instrument Panel (IP) Console

Geometry cleanup, casting-based mid-mesh extraction and dog house (clip/boss) capture using ANSA

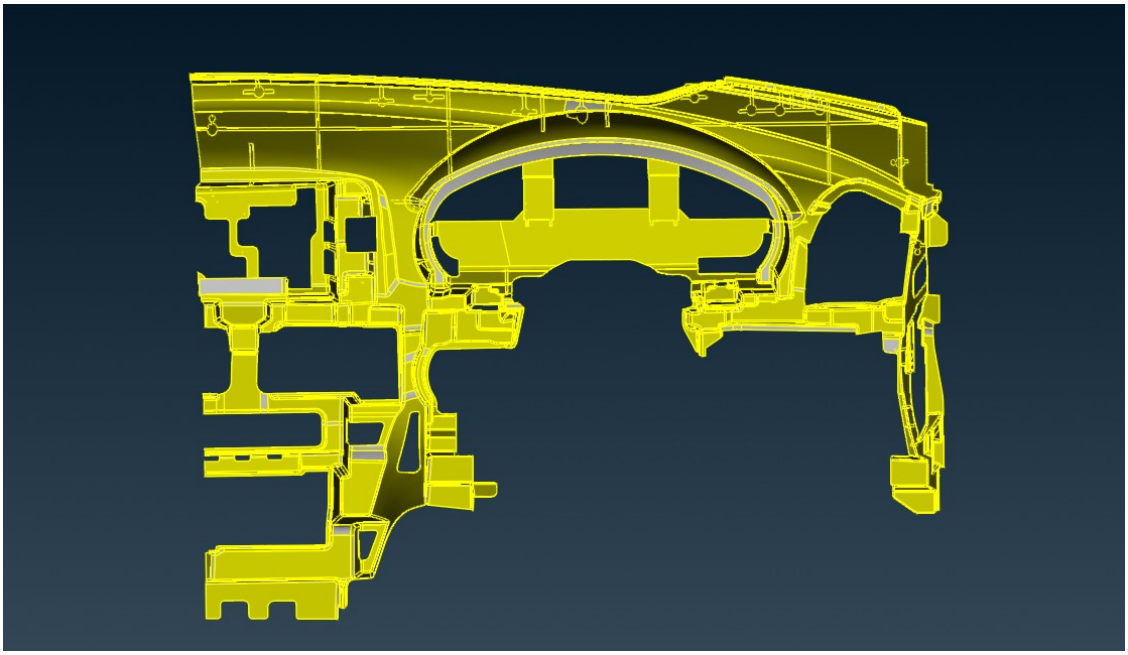


Figure 1: Instrument panel (IP) console – imported CAD geometry in ANSA

Item	Details
Component	Car IP console / dashboard (instrument cluster area, music-system area, glove-box side and full dashboard)
Material type	Plastic (injection moulded) – variable wall thickness
Pre-processor	ANSA (Geometry Checks, Mesh, Align Manager, Casting)
Method	Casting method for mid-mesh generation (mid-mesh, not mid-surface)
Prepared by	[Your name]
Date	October 2026

1. Objective and Scope

The objective of this project was to prepare a clean, analysis-ready finite element mid-mesh of a plastic car IP console (dashboard). The model includes the instrument cluster area (steering wheel and speedometer side), the music-player/infotainment area and the complete dashboard structure.

Because the part is moulded plastic with a wall thickness that changes from region to region, a mid-mesh (a shell mesh placed at the middle of the wall) was generated using the casting method instead of building a constant-thickness mid-surface. The scope of work covered:

- Importing the CAD (IGES) and running geometry checks
- Cleaning geometry errors (cracks, overlaps, needle faces, triple/single cons, problematic surfaces)
- Generating the mid-mesh with the Casting method
- Capturing the dog houses (clip/boss features) that were not captured properly by the automatic method
- Aligning perimeters, areas and imprints using Align Manager and removing white lines
- Fixing mesh holes/gaps and improving mesh quality

2. Why Mid-Mesh for a Plastic Part

Plastic components such as an IP console are thin-walled. Using solid elements would be very heavy, so the industry practice (BMW, Mercedes, VW, Audi and their Tier-1 suppliers) is to represent the wall with shell elements placed at the mid-plane of the wall, with the thickness stored as a property.

Mid-surface vs mid-mesh: A mid-surface is created in CAD-like geometry and then meshed. This works well for constant-thickness sheet metal. For plastic parts with ribs, bosses and varying thickness, the geometry-based approach is slow and error-prone. In the casting (mid-mesh) method, ANSA calculates the mid-plane directly from the solid and generates the shell mesh and the thickness together, which is faster and captures the variable thickness better.

3. Input Geometry and Geometry Checks

The part was imported as an IGES file (Dash Board Sub Parts-Final.IGS). The first step was to run the ANSA Checks Manager (Geometry) to find all geometrical problems. The check found a total of 371 errors.

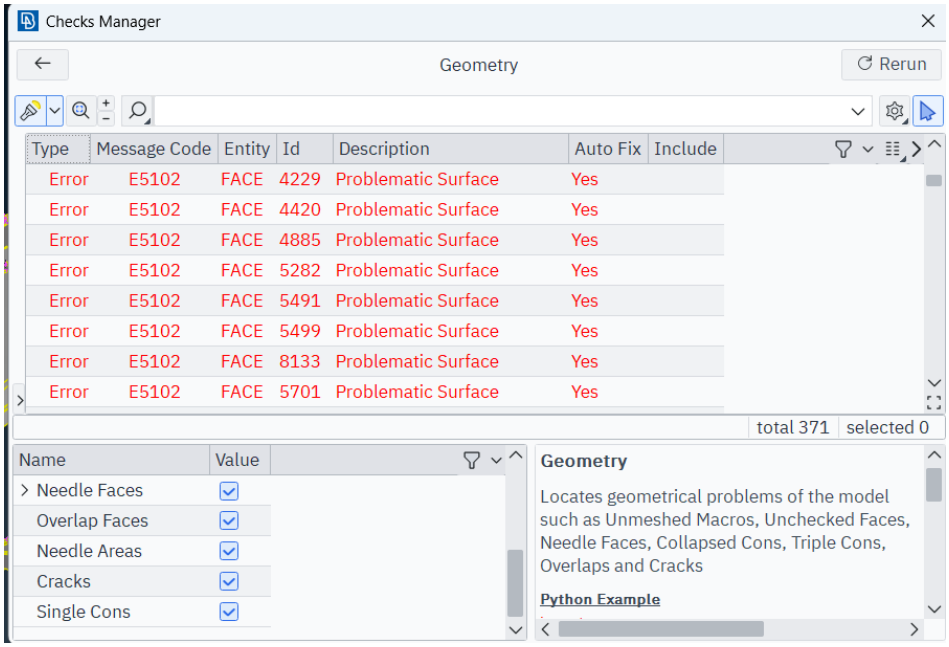


Figure 2: ANSA Checks Manager – Geometry check, total 371 errors found

3.1 Types of errors found

Code	Error	What it means (simple terms)
E5102	Problematic Surface	A surface whose shape is mathematically bad (twisted or degenerate) so meshing fails or gives poor elements.
E4068	Unchecked Faces	Faces the checker could not evaluate (26 faces).
E4071	Needle Face	A very thin, sliver-like face (e.g. face 342, 3231) that gives distorted elements.
E4072	Overlapping Faces	Two faces occupying the same space, causing duplicate/overlapping mesh.
E4073	Crack Between Cons	A tiny gap between neighbouring faces, so the mesh will not be continuous.
E4070	Triple Cons	Three faces sharing one edge – the model is not clean topologically.
E4074	Single Cons	A free edge with only one face; should be connected or deleted.

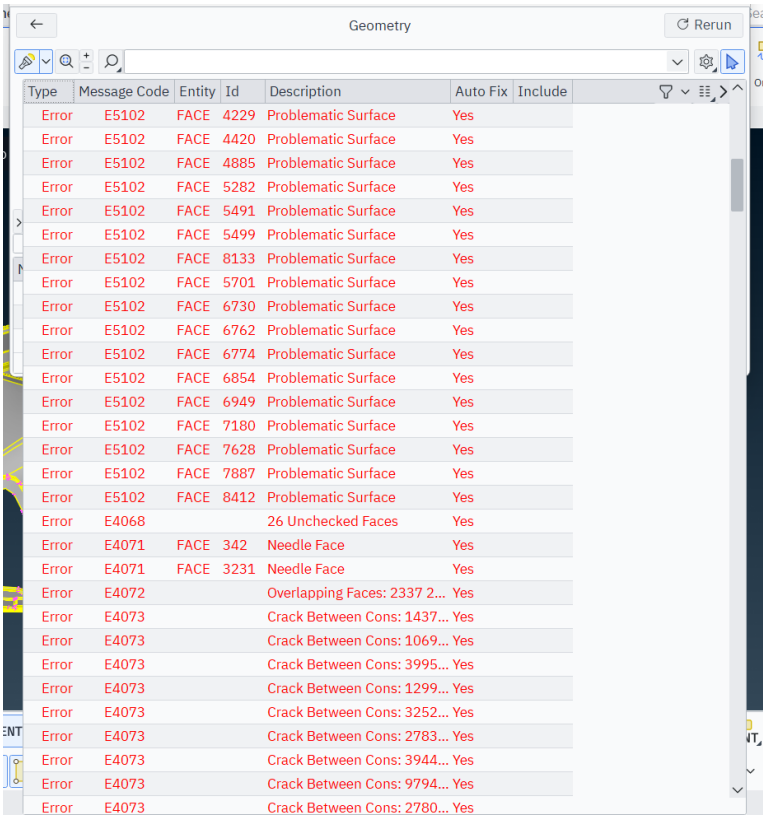


Figure 3: Geometry error list – Problematic Surface, Needle Face, Overlapping Faces and Cracks

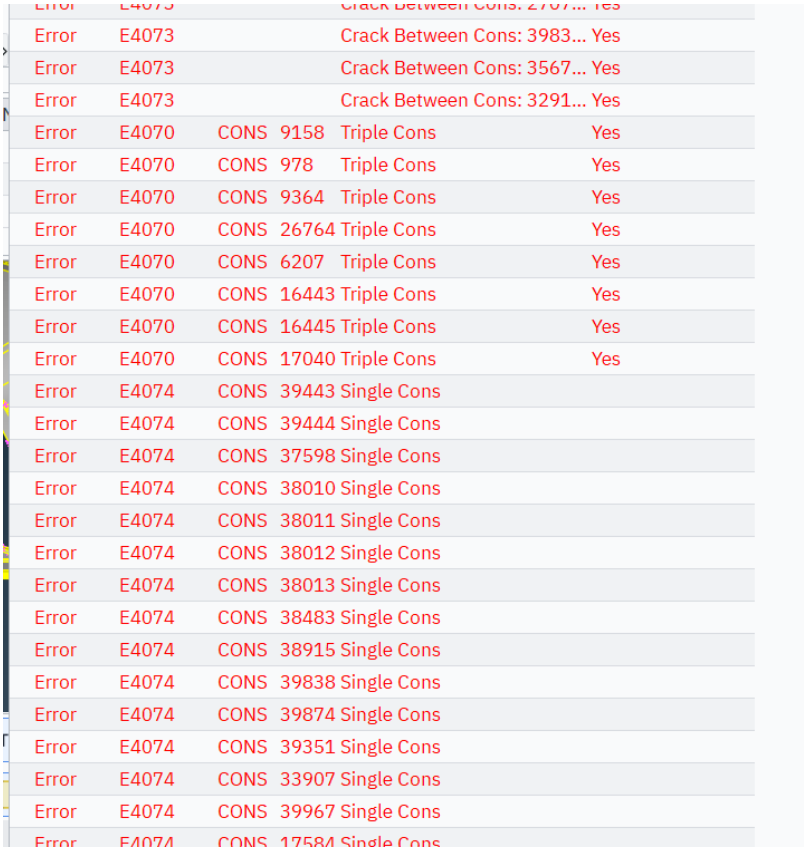


Figure 4: Geometry error list – Triple Cons and Single Cons

3.2 Cleanup actions performed

- **Auto-fix:** used the ANSA auto-fix for the errors marked “Yes” (cracks, needle faces, overlaps, triple cons).

- **Cons to Curve:** converted problematic constraint edges into curves so that the boundary could be rebuilt cleanly.
- **Curve editing:** Create, Connect, Modify, Project and Delete commands were used to rebuild or correct the curves.
- **Fill / Faces:** new faces were created with the Fill command to close open regions after the edges were corrected.
- **Single Cons:** remaining free edges were joined or removed so that the geometry became watertight.
- **Re-run of checks:** the Checks Manager was re-run after every set of fixes to confirm the model was clean.

4. Mid-Mesh Generation (Casting Method)

The casting method was used for the mid-mesh generation of the plastic IP console. The parameters below were applied.

Parameter	Value	Remark
Method	Casting	Mid-mesh (not mid-surface)
Minimum thickness	0.5 mm	Walls thinner than this are not extracted
Target length	4 mm	Average element size
Minimum target length	2 mm	Smallest allowed element size
Maximum target length	6 mm	Largest allowed element size
Thickness result (T result)	Taken / treated	Variable thickness stored on the mesh

After the extraction, the mid-mesh was reviewed region by region. Areas where the automatic method did not capture the geometry correctly – mainly the dog houses – were corrected manually as described in Section 5.

5. Dog House Capture

A dog house is a small moulded feature (clip, boss or rib pocket) attached to the wall of the dashboard. These features are thin and curved, so the casting method often misses part of them or leaves holes and gaps. Each missing dog house was captured manually with the workflow below.

Cross-pattern capture: The dog houses in this model are solid features that are hollow from the inside and do not carry a wall thickness of their own. Because of this, they were not meshed as a normal wall. Each dog house was captured in the mid-mesh as a cross pattern, where the internal walls of the dog house are represented by crossing shell faces connected to the main wall. This is why the curves and faces had to be created manually.

5.1 Workflow

1. Identify the dog house in the geometry and see how it is connected to the main wall.
2. Create curves along its edges using Cons to Curve (plus Connect, Create, Modify, Project and Delete where needed).

3. Build the missing faces with the Fill command.
4. Align perimeters and areas to the surrounding mid-mesh with Align Manager.
5. Fill remaining gaps or holes with FEM Topo.
6. Check connectivity and quality, then freeze the finished area.

5.2 Examples

Example 1 – cylindrical dog house: the CAD geometry and the mid-mesh generated around it.

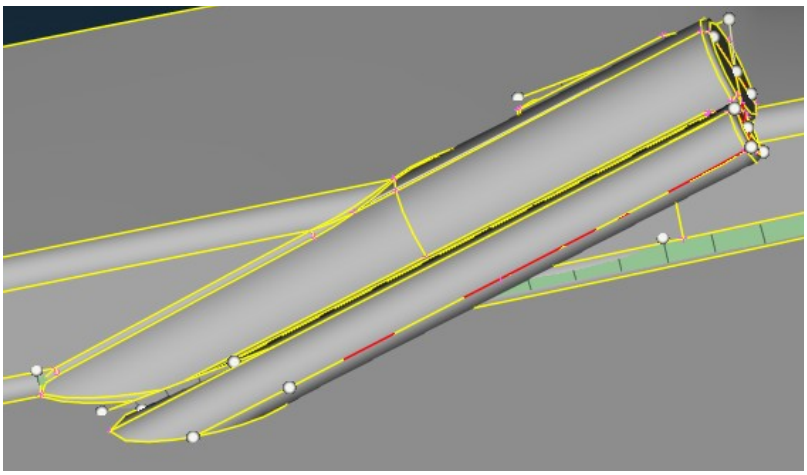


Figure 5: Dog house – CAD geometry with edges (yellow)

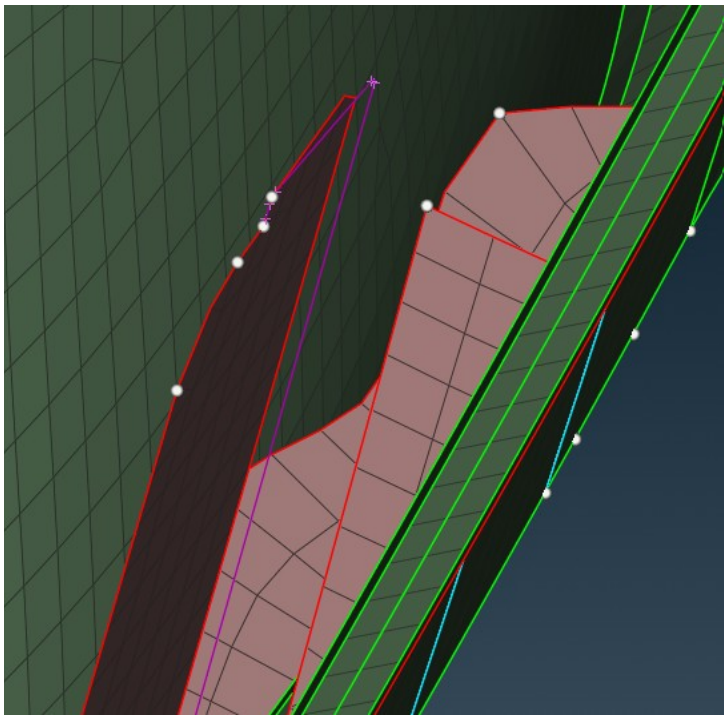


Figure 6: Dog house – mid-mesh before manual correction (gap visible)

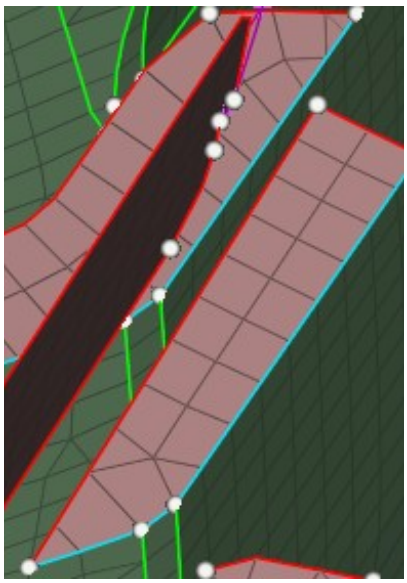


Figure 7: Dog house – mid-mesh gap along the dog house

Example 2 – dog house attached to the wall: the mesh was extended and attached to the wall mesh using Fill and Align.

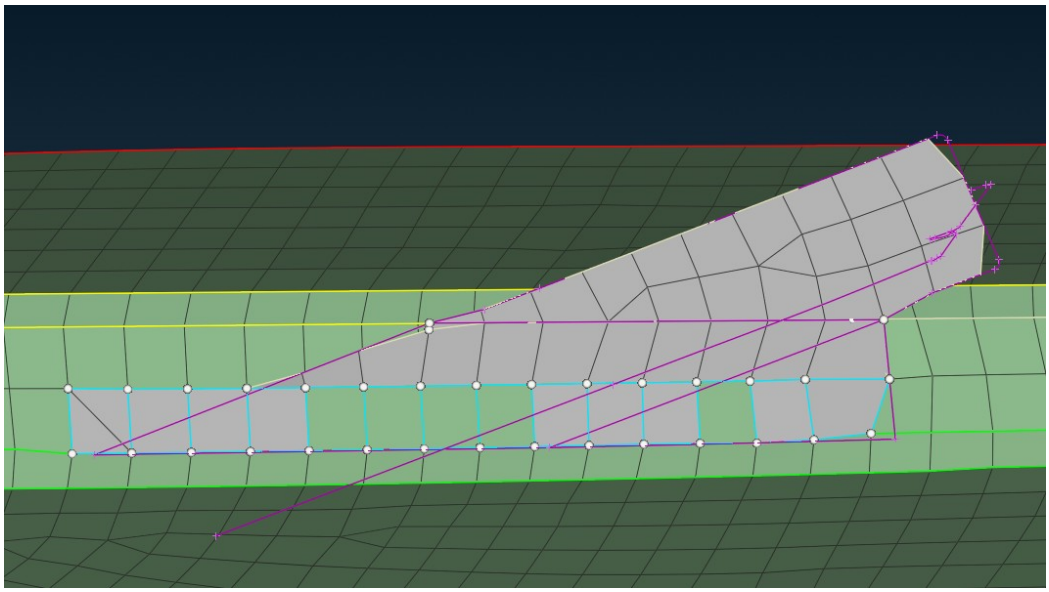


Figure 8: Dog house – mid-mesh attached to the wall using aligned perimeters

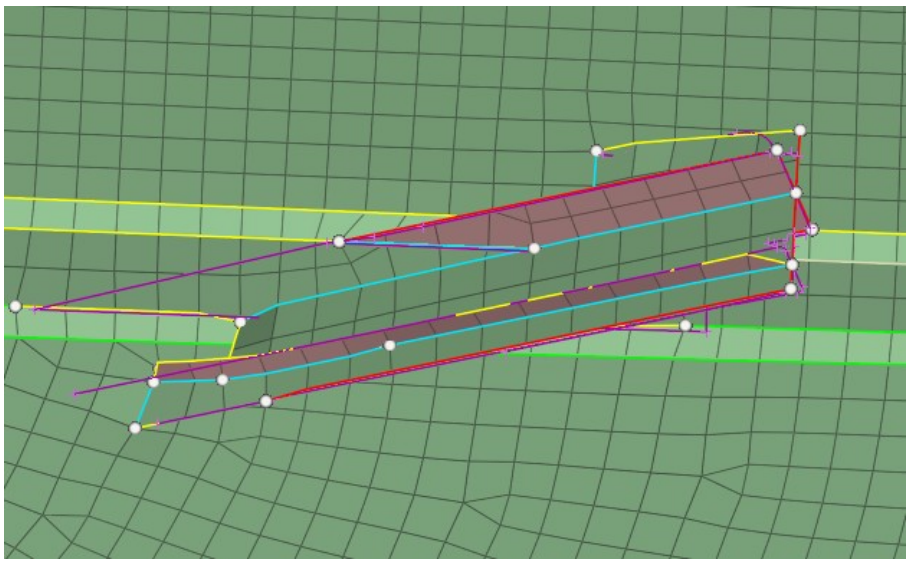


Figure 9: Dog house – attached faces and curves after Cons to Curve and Fill

Example 3 – quality check of a dog house. The coloured strip shows a mesh quality plot (red = worst, green = good) used to find and improve distorted elements.

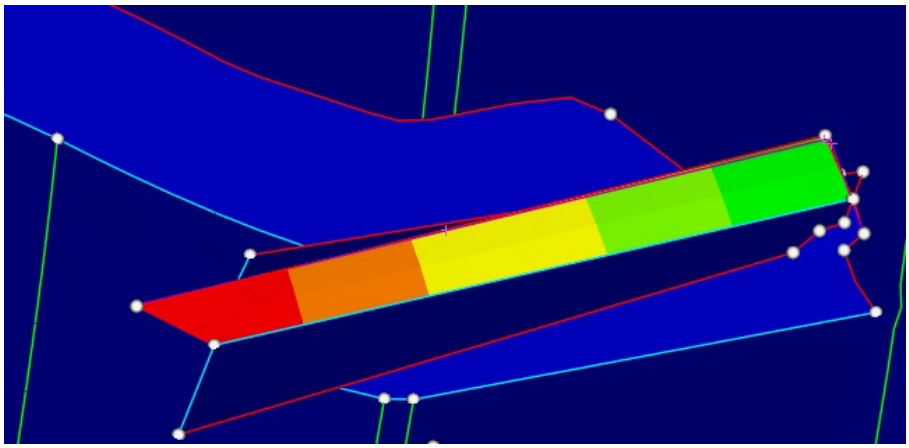


Figure 10: Dog house – element quality (analytical) plot

Example 4 – joining dog houses to the main wall.

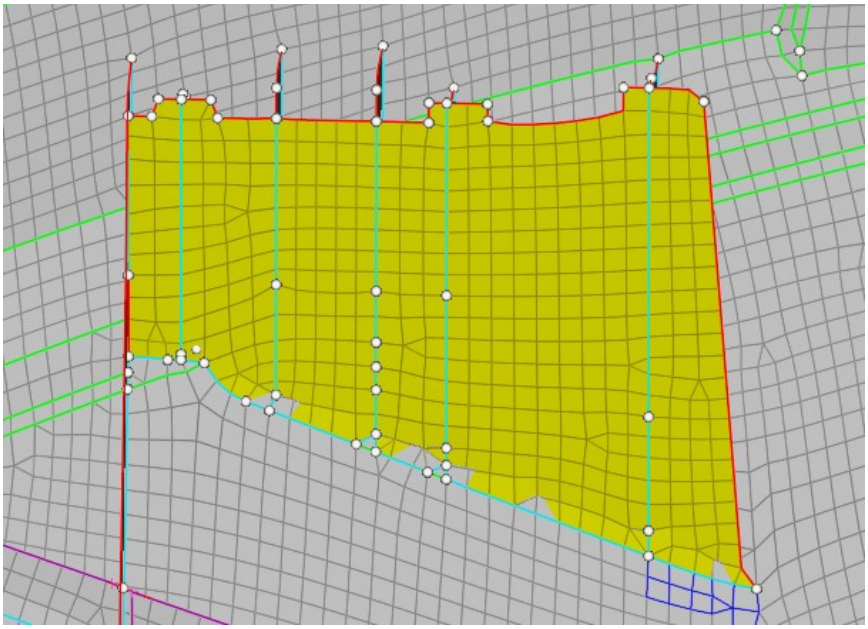


Figure 11: Dog house – joined with the surrounding mesh; blue lines show the aligned perimeter

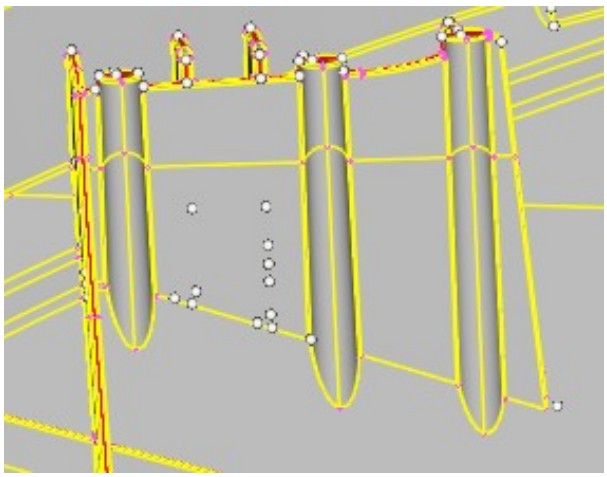


Figure 12: Group of three cylindrical dog houses

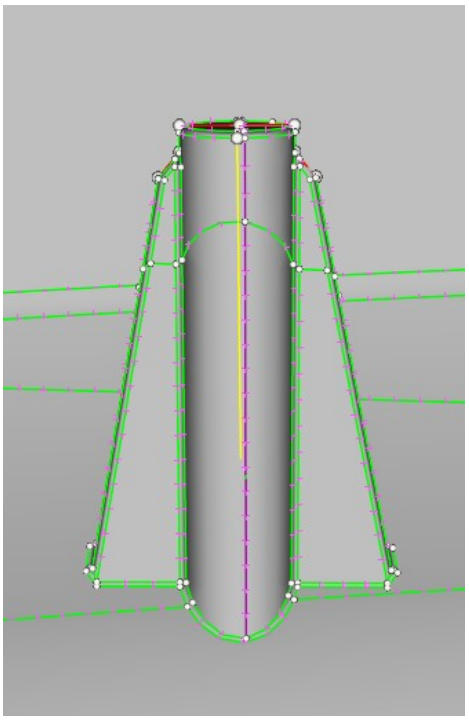


Figure 13: Dog house – tapered dog house with edges and perimeters

6. Align Manager, Imprinting and White-Line Removal

Align Manager was used extensively over the entire IP console. Its purpose is to make the mesh edges of neighbouring areas match each other so that the mesh is continuous with no gaps, overlaps or free edges.

- Perimeter alignment: edges of two neighbouring mesh areas are moved onto one common line.
- Area alignment: mesh area is adjusted to follow the curves and imprints of the geometry.
- Imprints: different imprint parameters were used for different areas of the console.
- White-line removal: white lines in the display mark free edges, which means the mesh is not connected. These were removed by aligning and connecting the edges.

Freeze and Unfreeze

When one area is aligned, the neighbouring area can get disturbed again and again. To avoid this, the finished area was frozen using the Freeze command. A frozen area does not change when a nearby

perimeter is modified, so the work done already stays safe. Unfreeze was used when that area needed to be edited again.

7. Commands and Tools Used

Command / tool	Purpose
Checks Manager (Geometry)	Finds cracks, overlaps, needle faces, triple/single cons and problematic surfaces
Cons to Curve	Converts face edges into curves to capture dog houses and repair geometry
Curves: Create / Connect / Modify / Project / Delete	Builds and edits curves to rebuild the boundaries
Faces / Fill	Creates new faces to close holes in geometry; in shell mesh it converts cons into mesh
Casting (mid-mesh)	Generates the variable-thickness mid-mesh from the plastic solid
Extrusion	Used for mid-surface generation in selected areas
Align Manager	Aligns perimeters, areas and imprints; removes white lines
Smooth / Reshape / Reconstruct	Improves element shape and quality
FEM Topo	Fills holes and gaps in the mid-mesh
Sweep	Creates elements by sweeping edges
Split (elements)	Splits elements for better layout and quality
Freeze / Unfreeze	Protects finished areas from being changed by later alignments

8. Final Mesh

Views of the finished mid-mesh of the full IP console are shown below. Pink/green areas are meshed shells, red lines are free boundaries of the part, green lines are connections between faces, and the small blue patches are interior features.

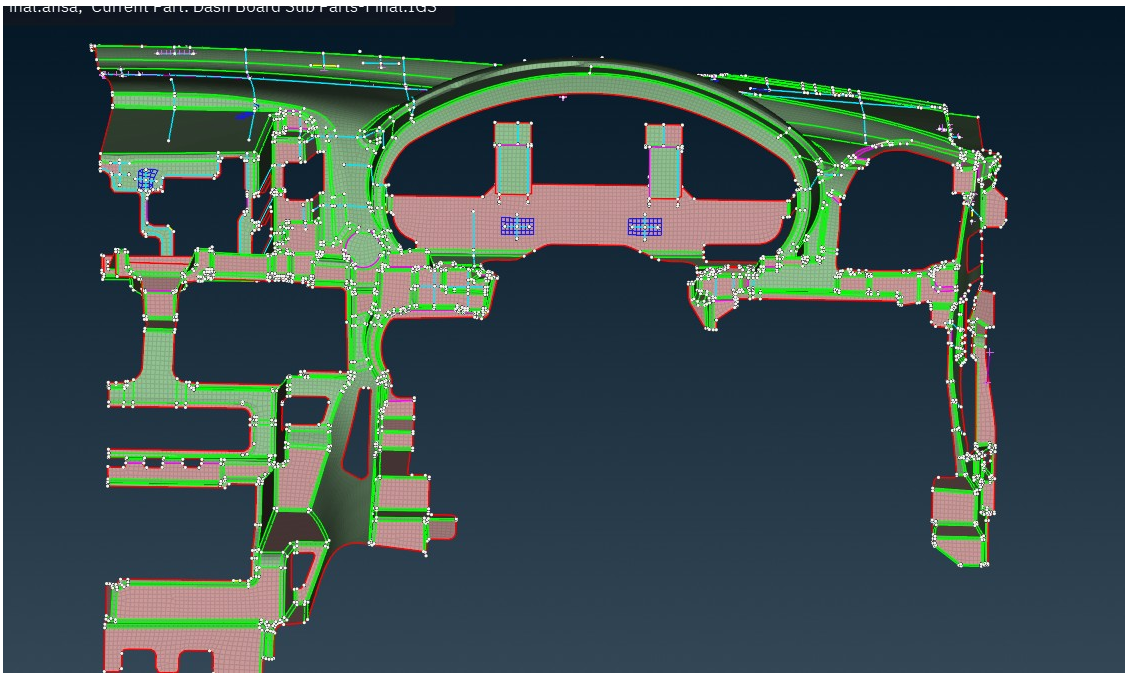


Figure 14: Final mid-mesh – front view

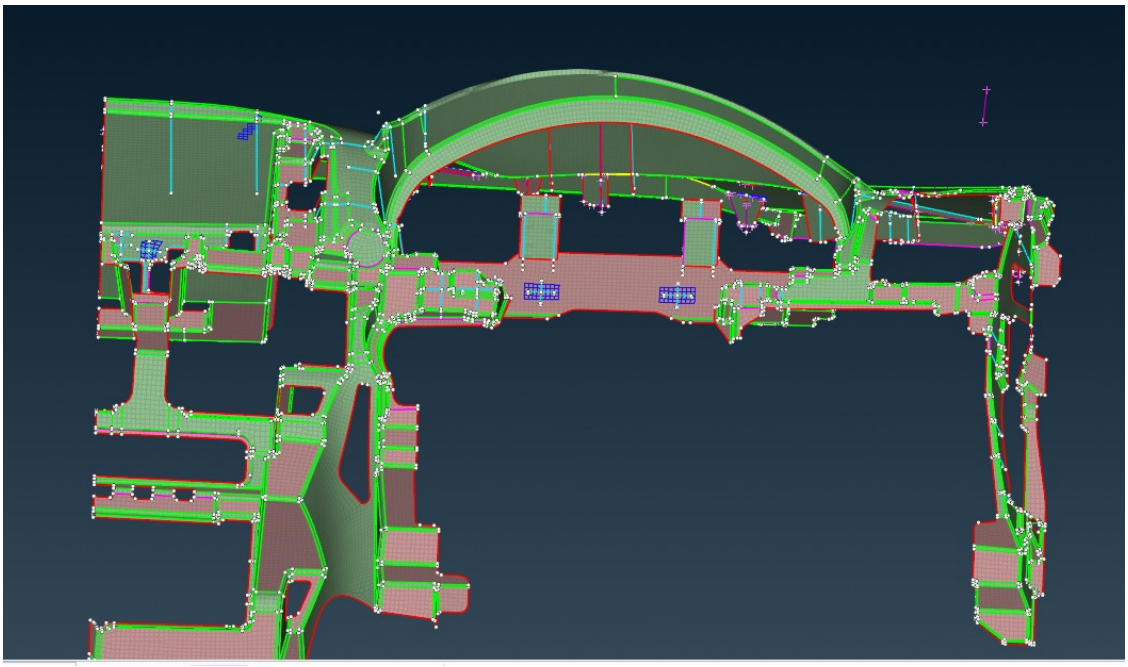


Figure 15: Final mid-mesh – view 2

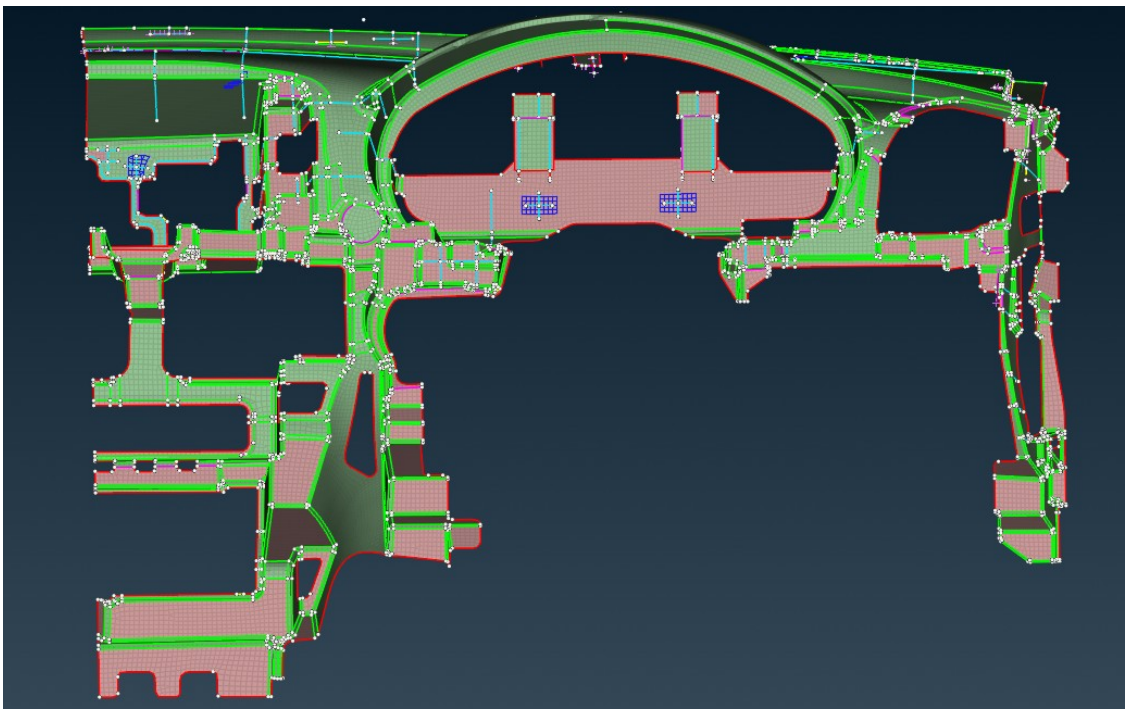


Figure 16: Final mid-mesh – view 3

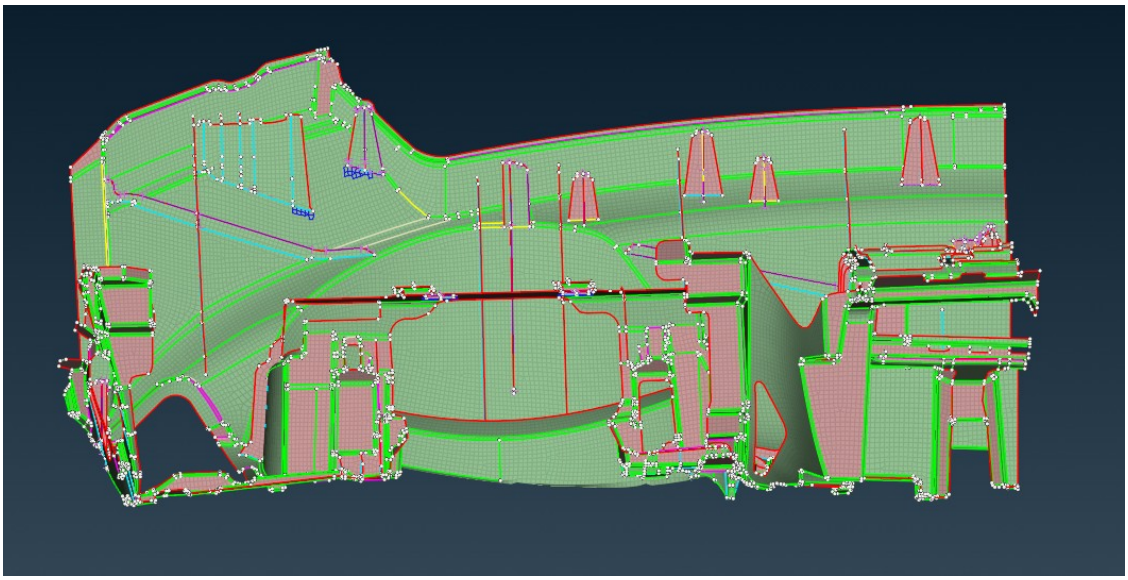


Figure 17: Final mid-mesh – top/side view

9. Mesh Statistics and Quality

9.1 Mesh statistics

The element count was taken from the ANSA Quality Monitor of the final model.

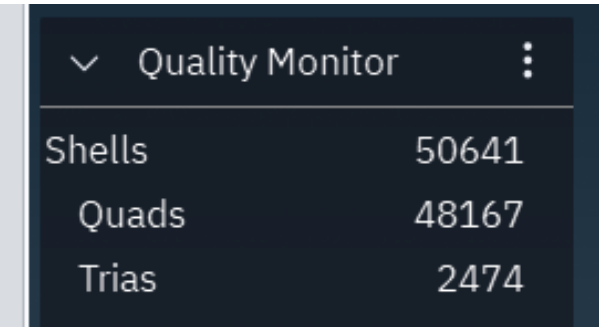


Figure 18: ANSA Quality Monitor – shell, quad and tria count of the final mid-mesh

Item	Result	Remark
Total number of shell elements	50,641	
Quad elements (QUAD4)	48,167 (95.11 %)	Dominant element type
Tria elements (TRIA3)	2,474 (4.89 %)	Well below the 15 % limit

The mesh is made of 95.11 % quads and only 4.89 % trias. The tria share is far below the allowed 15 %, which indicates a good, regular mesh flow over the console.

9.2 Quality criteria

The quality criteria used for this model are listed below. The “Result” and “Status” columns show how the final mesh compares with each limit.

S.No.	Quality criterion	Required value	Result	Status
1	Target / average length	4 mm	4 mm (set)	Pass
2	Minimum length	2 mm	2 mm (set)	Pass
3	Maximum length	6 mm	6 mm (set)	Pass
4	Aspect ratio	≤ 3	Within limit	Pass
5	Warpage	≤ 15°	Within limit	Pass
6	Skewness	≤ 45°	Within limit	Pass
7	Jacobian	≥ 0.7	Within limit	Pass
8	Minimum quad angle	≥ 45°	Within limit	Pass
9	Maximum quad angle	≤ 135°	Within limit	Pass
10	Minimum tria angle	≥ 30°	Within limit	Pass
11	Maximum tria angle	≤ 120°	Within limit	Pass
12	Tria percentage	≤ 15 %	4.89 %	Pass

Note: a few failed elements can remain near triple cons or in very small regions bounded by two triple cons. These are normally kept as they are when they cannot be solved; all other elements must meet the criteria above.

10. Conclusion

A clean mid-mesh of the plastic IP console was generated using the casting method with a minimum thickness of 0.5 mm and element sizes between 2 mm and 6 mm (target 4 mm). Geometry errors found in the Checks Manager were cleaned using Cons to Curve, curve editing, Fill and auto-fix. Dog houses that were not captured correctly were rebuilt manually, and Align Manager, Freeze/Unfreeze, FEM Topo, Smooth, Reshape and Reconstruct were used to obtain a continuous mesh without white lines, gaps or holes. The final model contains 50,641 shell elements, of which 48,167 are quads (95.11 %) and 2,474 are trias (4.89 %), well below the 15 % tria limit. The resulting model is suitable for use in downstream structural, NVH or crash analysis.

Skills demonstrated

- Geometry cleanup and topology repair in ANSA
- Variable-thickness mid-mesh generation for plastic parts
- Manual capture of complex features (dog houses, clips, bosses)
- Mesh quality control to OEM-style criteria